COURSE 9 – IBM DATA ANALYST CAPSTONE PROJECT

MODULE 1: DATA COLLECTION

1.1 COURSE INTRODUCTION

The IBM Data Analyst Capstone Project is designed to help learners apply all the concepts and tools learned throughout the IBM Data Analyst Professional Certificate. In this course, learners work as an Associate Data Analyst and solve real-world business problems using industry-standard analytics tools.

The course focuses on:

- Data collection
- Data wrangling
- Exploratory Data Analysis (EDA)
- Data visualization
- Dashboard creation
- Presenting insights professionally

The course uses practical datasets and business scenarios to strengthen analytical thinking and technical skills.

1.2 IMPORTANCE OF DATA ANALYTICS

The demand for data analysts is growing rapidly because organizations increasingly rely on data-driven decision-making.

Data analysts help organizations:

- Interpret large volumes of data
- Identify trends and patterns
- Support strategic decisions

- Improve operational efficiency
- Predict future outcomes

This capstone course helps learners gain hands-on experience in solving real business problems using data.

1.3 REQUIRED PREREQUISITES

Before starting this capstone project, learners should already know:

Technical Skills

- Python programming
- Jupyter Notebooks
- SQL
- Relational databases
- Data wrangling
- Data analysis
- Data visualization
- Dashboarding tools

Familiarity with Libraries and Tools

- Pandas
- NumPy
- Matplotlib
- Seaborn
- Plotly
- Dash
- IBM Cognos Analytics or Google Looker

1.4 Course Objectives

By the end of the course, learners will be able to:

- Collect data using APIs and web scraping
- Explore and clean datasets
- Handle duplicates and missing values
- Normalize and prepare data
- Perform Exploratory Data Analysis (EDA)
- Analyze distributions and correlations
- Create meaningful visualizations
- Build dashboards using BI tools
- Present findings professionally

Course Structure

Module 1 – Data Collection

Topics covered:

- Collecting data using APIs
- Collecting data using web scraping
- Exploring datasets

Module 2 – Data Wrangling

Topics covered:

- Finding duplicates
- Removing duplicates
- Finding missing values
- Imputing missing values
- Normalizing data

Module 3 – Exploratory Data Analysis (EDA)

Topics covered:

- Analyzing data distributions
- Handling outliers
- Correlation analysis

Module 4 – Data Visualization

Topics covered:

- Visualizing distributions
- Visualizing relationships
- Visualizing compositions
- Visualizing comparisons

Module 5 – Dashboard Building

Topics covered:

- Creating dashboards
- Organizing visualizations
- Presenting insights interactively

Module 6 – Final Presentation

Topics covered:

- Presenting findings
- Report structure
- Storytelling with data
- Best presentation practices

1.5 EMERGING TRENDS IN DATA ANALYTICS

Modern organizations increasingly rely on advanced analytics technologies to gain real-time insights and maintain competitiveness.

The major emerging trends are:

1. AI and Machine Learning Integration
2. Data Governance and Ethics
3. Real-Time Data Processing
4. Augmented Analytics
5. Cloud-Native Analytics
6. Enhanced Data Visualization
7. Predictive and Prescriptive Analytics

1.6 AI AND MACHINE LEARNING INTEGRATION

Artificial Intelligence (AI) and Machine Learning (ML) are becoming deeply integrated into analytics workflows.

Key Areas

- Natural Language Processing (NLP)
- Computer Vision
- Predictive Analytics
- Automated Machine Learning (AutoML)

Benefits

- Faster insights
- Improved prediction accuracy
- Scalable analytics systems
- Easier model creation for non-specialists

Business Impact

Organizations use AI-driven analytics to automate decision-making and improve efficiency.

1.7 DATA GOVERNANCE AND ETHICS

As data privacy concerns increase, organizations are strengthening data governance frameworks.

Important Regulations

- GDPR (General Data Protection Regulation)
- CCPA (California Consumer Privacy Act)

Key Focus Areas

- Data privacy
- Transparency
- Responsible AI
- Ethical data handling
- Bias prevention

Importance

Proper governance helps maintain trust and compliance.

1.8 REAL-TIME DATA PROCESSING AND STREAMING ANALYTICS

Organizations increasingly require real-time insights.

Technologies Used

- Apache Kafka
- Spark Streaming
- Apache Flink

Applications

- Finance
- Healthcare

- Retail
- Logistics

Benefits

- Immediate decision-making
- Faster response to events
- Operational efficiency

1.9 AUGMENTED ANALYTICS

Augmented analytics uses AI and ML to automate many analytics tasks.

Automated Tasks

- Data preparation
- Insight discovery
- Data explanation

Advantages

- Reduces manual work
- Increases productivity
- Helps non-technical users analyze data
- Accelerates decision-making

1.10 CLOUD-NATIVE DATA ANALYTICS

Cloud platforms are becoming the standard for analytics.

Major Cloud Providers

- Amazon Web Services (AWS)
- Microsoft Azure
- Google Cloud Platform (GCP)

Benefits

- Scalability

- Cost efficiency
- Easy integration
- Reduced infrastructure management

Cloud analytics allows businesses to perform advanced analysis without large on-premises systems.

1.11 ENHANCED DATA VISUALIZATION AND STORYTELLING

Visualization techniques are evolving beyond static charts.

Modern Visualization Features

- Interactive dashboards
- Dynamic charts
- Immersive visuals
- Storytelling dashboards

Importance

Good visualizations help stakeholders:

- Understand data quickly
- Identify trends easily
- Make informed decisions

1.12 PREDICTIVE AND PRESCRIPTIVE ANALYTICS

Advanced analytics techniques are widely used for forecasting and optimization.

Predictive Analytics

Predicts future outcomes based on historical data.

Examples:

- Sales forecasting
- Demand prediction
- Risk analysis

Prescriptive Analytics

Recommends optimal actions.

Examples:

- Inventory optimization
- Pricing strategies
- Route optimization

Industries Benefiting

- Healthcare
- Finance
- Logistics
- Retail

1.13 CAPSTONE PROJECT OVERVIEW

In this capstone project, learners act as Associate Data Analysts at a technology consulting company.

The main objective is to analyze technology trends using the latest Stack Overflow Developer Survey dataset.

The insights generated will help organizations make strategic decisions.

1.13.1 Main Goals of the Project

The project focuses on analyzing:

- Programming language trends
- Database technology trends
- Platform trends
- Framework trends
- Developer preferences
- Future learning interests

The project identifies:

- Current technology usage
- Future technology trends
- Emerging skill demands

1.13.2 Tools and Technologies Used

Programming and Analysis

- Python
- SQL
- Pandas
- NumPy

Visualization

- Matplotlib
- Seaborn
- Plotly

Dashboarding

- IBM Cognos Analytics
- Google Looker Studio

Development Environment

- Jupyter Notebook

1.13.3 Dataset Used in the Project

The project uses a filtered version of the Stack Overflow Developer Survey dataset.

The dataset contains:

- Technologies developers currently use
- Technologies developers want to learn

- Developer preferences
- Industry trends

The data is preprocessed for consistency.

1.13.4 Project Workflow

The project follows a complete data analytics pipeline.

Step 1 – Data Collection

Data is collected using:

- APIs
- Web scraping
- CSV files
- Excel files
- Databases

Sources of Data

- Job postings
- Training portals
- Developer surveys

1.14 DATA COLLECTION USING APIS

APIs allow analysts to retrieve structured data programmatically.

Key Concepts

- HTTP requests
- Endpoints
- JSON responses
- Authentication

Skills Learned

- Sending API requests

- Accessing API data
- Parsing JSON
- Storing retrieved data

1.15 DATA COLLECTION USING WEB SCRAPING

Web scraping extracts information from websites.

Common Tools

- BeautifulSoup
- Requests
- Pandas

Skills Learned

- Extracting HTML data
- Parsing tables
- Cleaning scraped data
- Automating data collection

1.16 DATA COLLECTION LABS

Lab 1: Review of Accessing APIs

Objectives

After completing this lab, you should be able to:

- Understand HTTP
- Analyze HTTP requests
- Understand GET and POST requests
- Work with request and response objects
- Read API responses in JSON format

Understanding HTTP

HTTP stands for:

HyperText Transfer Protocol

It is the communication protocol used between:

- Clients (browser/Python program)
- Servers (websites/APIs)

Types of HTTP Requests

The lab mainly focuses on:

1. GET Request
2. POST Request

GET Request

A GET request is used to:

- Retrieve data from a server
- Pass parameters through the URL

Example URL

```
r.url
```

Output:

```
http://httpbin.org/get?name=Joseph&ID=123
```

Observation

- Parameters appear in the URL
- No request body exists

POST Request

A POST request is used to:

- Send data to the server
- Submit forms or data securely

Example

```
response.url
```

Output:

```
http://httpbin.org/post
```

Observation

- Parameters are NOT visible in URL
- Data is sent in the request body

Comparing GET vs POST

Comparing URLs

```
print("POST request URL:",response.url )
print("GET request URL:",r.url)
```

Output:

```
POST request URL: http://httpbin.org/post
GET request URL: http://httpbin.org/get?name=Joseph&ID=123
```

Key Difference

GET	POST
Parameters visible in URL	Parameters hidden
Used for fetching data	Used for sending data

Comparing Request Bodies

```
print("POST request body:",response.request.body)
print("GET request body:",r.request.body)
```

Output:

```
POST request body: name=Joseph&ID=123
GET request body: None
```

Important Observation

POST Request

- Contains a request body
- Sends data securely

GET Request

- No request body

- Data travels through URL

Viewing Form Data

```
response.json()['form']
```

Output:

```
{'ID': '123', 'name': 'Joseph'}
```

Explanation

The server returns data in JSON format.

The .json() method converts response data into a Python dictionary.

Important API Concepts

Request

Data sent from client to server.

Response

Data returned by server.

JSON

A lightweight data-interchange format commonly used in APIs.

Example:

```
{
 "ID": "123",
 "name": "Joseph"
```

Key Learnings from Lab 1

- HTTP enables communication between clients and servers
- GET requests retrieve data
- POST requests send data
- GET stores parameters in URL
- POST stores parameters in request body
- APIs commonly return JSON responses
- Python Requests library simplifies API access

Lab 2: Collecting Data Using APIs

Objectives

After completing this lab, you should be able to:

- Access APIs using Python
- Retrieve job posting data
- Store API data in Excel files
- Work with structured datasets

Scenario

The task is to collect the number of job postings for different technologies using an API.

Technologies Analyzed

```
skills = [
    "C",
    "C#",
    "C++",
    "Java",
    "JavaScript",
    "Python",
    "Scala",
    "Oracle",
    "SQL Server",
    "MySQL Server",
    "PostgreSQL",
    "MongoDB"
]
```

Technologies Included

Programming Languages

- C
- C#
- C++
- Java

- JavaScript
- Python
- Scala

Databases

- Oracle
- SQL Server
- MySQL Server
- PostgreSQL
- MongoDB

Writing Data into Excel

Adding Headers

```
ws.append(["Technology", "Number of Jobs"])
```

This creates column names in the worksheet.

Output Columns

```
| Technology | Number of Jobs |
```

Fetching Job Counts

```
for skill in skills:
    number_of_jobs = get_number_of_jobs_T(skill)[1]
    ws.append([skill, number_of_jobs])
```

Explanation of Workflow

Step 1

Loop through each technology.

Step 2

Call API function:

```
get_number_of_jobs_T(skill)
```

Step 3

Retrieve job count.

Step 4

Store results in worksheet.

Saving Workbook

```
wb.save("skill-job-postings.xlsx")
```

Result

An Excel file is created:

```
skill-job-postings.xlsx
```

Key Learnings from Lab 2

- APIs can retrieve real-world job data
- Python automates data collection
- Excel files can store structured results
- Loops help automate repetitive tasks
- APIs are useful for labor market analysis

Lab 3: Review of Web Scraping

Objectives

After completing this lab, you should be able to:

- Understand web scraping
- Extract HTML table data
- Parse webpages using BeautifulSoup
- Retrieve specific elements from HTML

What is Web Scraping?

Web scraping means:

Extracting information from websites automatically using code.

Important HTML Tags Used

HTML Tag	Purpose
<table>	Table
<tr>	Table row
<td>	Table column/data

Extracting Table Rows

```
for row in table.find_all('tr'):
```

Explanation

- find_all('tr') finds all rows in table

Extracting Table Columns

```
cols = row.find_all('td')
```

Explanation

- Finds all columns inside a row

Extracting Specific Columns

```
color_name = cols[2].getText()
color_code = cols[3].getText()
```

Explanation

cols[2]

Third column → Color Name

cols[3]

Fourth column → Hex Code

Printing Data

```
print("{}--->{}".format(color_name,color_code))
```

Sample Output

```
lightsalmon--->#FFA07A
salmon--->#FA8072
gold--->#FFD700
orange--->#FFA500
green--->#008000
dodgerblue--->#1E90FF
```

Understanding Hex Codes

Hex color codes represent colors digitally.

Example:

Color	Hex Code
Red	#FF0000
Green	#00FF00
Blue	#0000FF

Key Learnings from Lab 3

- BeautifulSoup helps parse HTML
- Tables can be scraped row by row
- HTML tags help locate content
- .getText() extracts readable text
- Web scraping automates data extraction

Lab 4: Collecting Data Using Web Scraping

Objectives

After completing this lab, you should be able to:

- Scrape real-world data from websites
- Extract salary information

- Store scraped data in CSV files
- Convert scraped data into DataFrames

Task

Scrape:

- Programming language names
- Average annual salaries

Extracting Table Data

```
table = soup.find('table')
```

This finds the first HTML table on webpage.

Loop Through Rows

```
for row in table.find_all('tr')[1:]:
```

Important Detail

[1:]

Skips the header row.

Extracting Columns

```
cols = row.find_all('td')
```

Extracting Language and Salary

```
language = cols[1].text
salary = cols[3].text
```

Displaying Output

```
print(language, salary)
```

Sample Output

Language	Salary
Python	$114,383
Java	$101,013

Language	Salary
R	$92,037
JavaScript	$110,981
Swift	$130,801
C++	$113,865
SQL	$84,793

Creating DataFrame

```
df = pd.DataFrame(columns=["Language", "Average Annual Salary"])
```

Adding Rows to DataFrame

```
df.loc[len(df)] = [language, salary]
```

Saving CSV File

```
df.to_csv("popular-languages.csv", index=False)
```

Result

CSV file created:

popular-languages.csv

Importance of CSV Files

CSV files are:

- Lightweight
- Easy to share
- Supported by Excel
- Common in data analysis

Key Learnings from Lab 4

- Web scraping can collect salary data
- BeautifulSoup extracts HTML content
- Pandas stores structured data

- CSV files are widely used in analytics
- Automation saves time during data collection

1.17 STACK OVERFLOW DEVELOPER SURVEY DATASET

The project mainly uses the Stack Overflow Developer Survey dataset.

Dataset Characteristics

Contains information about:

- Developers worldwide
- Technologies used
- Career information
- Preferences
- Salaries
- Learning methods

Important Dataset Columns

Column	Description
Age	Age group
Employment	Employment status
RemoteWork	Remote working frequency
YearsCode	Coding experience
YearsCodePro	Professional coding experience
DevType	Developer role
Country	Country of residence
CompTotal	Total compensation
LanguageHaveWorkedWith	Languages used

Column	Description
LanguageWantToWorkWith	Languages desired
DatabaseHaveWorkedWith	Databases used
PlatformHaveWorkedWith	Platforms used
WebframeHaveWorkedWith	Web frameworks used
AISelect	Opinion about AI tools
JobSat	Job satisfaction

1.18 EXPLORING THE DATASET

Exploratory Data Analysis (EDA) begins with understanding the dataset structure.

Mapping Age Categories

The dataset stores age as ranges.

Example:

```
age_mapping = {
    "18-24 years old": 21,
    "25-34 years old": 29
}
```

Creating Numeric Age Column

```
df['Age_numeric'] = df['Age'].map(age_mapping)
```

Calculating Mean Age

```
mean_age = df['Age_numeric'].mean()
```

Result:

```
32.68
```

Counting Unique Countries

```
df['Country'].nunique()
```

Output:

```
185
```

This indicates respondents came from 185 countries.

KEY LEARNINGS FROM MODULE 1

By the end of Module 1, you should understand:

APIs

- GET requests
- POST requests
- Request bodies
- JSON responses

Web Scraping

- HTML parsing
- Extracting tables
- Using BeautifulSoup

Data Collection

- APIs
- Scraping
- CSV generation
- Excel generation

Dataset Exploration

- Understanding columns
- Mapping categorical data
- Basic statistics
- Unique value analysis

Important Python Libraries Used

Library	**Purpose**
pandas	Data analysis
numpy	Numerical operations
requests	API requests
BeautifulSoup	Web scraping
openpyxl	Excel handling

MODULE 2: DATA WRANGLING

2.1 OVERVIEW OF DATA WRANGLING

Data wrangling (or **data munging**) is the process of:

- Cleaning raw data
- Removing inconsistencies
- Handling missing values
- Transforming data into usable formats
- Preparing dataset for analysis

Why Data Wrangling is Important

- Ensures data accuracy
- Prevents misleading results
- Improves analysis quality
- Makes dataset machine-readable

Dataset Used

The **Stack Overflow Developer Survey dataset** is used.

It contains:

- Developer demographics
- Technology usage
- Salary information
- Experience details
- Employment patterns

2.2 FINDING DUPLICATES

Objective

Identify and remove duplicate rows to maintain **data integrity**.

Why duplicates are a problem

- Inflate dataset size
- Skew analysis results
- Bias statistical calculations
- Lead to incorrect insights

Types of duplicates

1. Exact duplicate rows

All column values are identical.

2. Partial duplicates

Some columns match but not all.

Strategy Used

Full-row comparison approach

A row is considered duplicate only if:

All column values are exactly the same

Why full-row comparison is used

- Survey responses often have similar answers across users

- Similar responses ≠ duplicate records
- Prevents accidental deletion of valid data

Role of Unique Identifier

- ResponseId is the best candidate for uniqueness
- However, duplicates were still checked using full row match

Key Insight

✓ Duplicate removal is safest when based on **complete row equality**

2.3 HANDLING MISSING VALUES

Objective

Identify missing values to evaluate data completeness.

Why missing values matter

- Reduce accuracy of analysis
- Affect statistical results
- Break machine learning models
- Distort distributions

Steps performed

1. Count missing values per column

Used to understand data quality issues.

2. Identify high-missing columns

Focus on important variables like:

- Compensation
- Employment
- Education

Common handling methods

- Mean imputation (numerical)
- Median imputation (skewed data)

- Mode imputation (categorical)

2.4 EMPLOYMENT STATUS ANALYSIS

Column Used:

Employment

Purpose

To understand:

- Distribution of job types
- Employment patterns among developers
- Workforce segmentation

Technique

- Value counts applied
- Frequency distribution analysis performed

Insight

This helps identify:

- Full-time vs part-time developers
- Freelancers vs employed professionals
- Global employment trends

2.5 DATA NORMALIZATION

Why normalization is needed

Different numeric scales can distort analysis.

Example:

- Salary ranges from 1,000 to 1,000,000
- Other metrics may be smaller

Columns used

- CompTotal

- ConvertedCompYearly

2.6 COMPENSATION ANALYSIS

Key Columns

Column	Meaning
CompTotal	Total compensation
ConvertedCompYearly	Annual converted salary

Issues observed

- Extremely large values
- Outliers present
- Skewed distribution

Summary Statistics Insight

Example:

- Mean is extremely high due to outliers
- Median is more reliable than mean
- Large standard deviation indicates spread

2.7 HANDLING MISSING COMPENSATION VALUES

Techniques used

1. Median imputation

- Used for skewed salary data
- More robust than mean

2. Missing value replacement

Ensures dataset completeness

2.8 NORMALIZATION TECHNIQUES

1. Min-Max Scaling

Formula:

$$X_{scaled} = \frac{X - X_{min}}{X_{max} - X_{min}}$$

Purpose:

- Scales values between 0 and 1
- Useful for comparison

2. Z-Score Normalization

Formula:

$$Z = \frac{X - \mu}{\sigma}$$

Purpose:

- Centers data around mean = 0
- Shows how far values are from average

Visualization Output

Histogram analysis:

- Min-Max shows uniform range (0–1)
- Z-score shows distribution around 0

2.9 MISSING VALUE IMPUTATION EXAMPLE

Categorical column example: Employment

Method used:

- Mode imputation (most frequent value)

Visualization after imputation

- Bar chart used
- Shows top employment categories
- Confirms distribution remains logical

2.10 DATA WRANGLING LAB SUMMARY

In this module, you learned how to:

1. Handle Duplicates

- Identify full-row duplicates
- Remove redundant records
- Maintain dataset integrity

2. Handle Missing Values

- Detect missing values
- Impute using mean/median/mode
- Validate completeness

3. Analyze Employment Data

- Value counts
- Distribution understanding
- Workforce segmentation

4. Normalize Data

- Min-Max scaling
- Z-score normalization
- Compare scaled distributions

5. Feature Engineering

- Create new variables (e.g., ExperienceLevel)
- Improve dataset usability

6. Transform Data

- Log transformations

- Skewness reduction
- Improved model readiness

KEY PYTHON TECHNIQUES USED

Technique	Library
Value counts	pandas
Missing value detection	pandas
Imputation	pandas
Scaling	sklearn / manual
Visualization	matplotlib

MODULE 3: EXPLORATORY DATA ANALYSIS (EDA)

3.1 INTRODUCTION TO EXPLORATORY DATA ANALYSIS (EDA)

Exploratory Data Analysis (EDA) is the process of analyzing datasets to understand their structure, identify patterns, detect anomalies, and discover relationships between variables before applying advanced analysis or machine learning.

In this module, the Stack Overflow Developer Survey dataset is analyzed after cleaning and preprocessing.

Objectives

After completing this module, you will be able to:

- Understand dataset structure and data types
- Analyze distributions of important variables
- Identify and handle missing values
- Detect and remove outliers

- Compute statistical measures such as:
    - Mean
    - Median
    - Interquartile Range (IQR)
    - Correlation
- Analyze relationships between variables
- Create visualizations using:
    - Histograms
    - Boxplots
    - Scatterplots
    - Heatmaps
- Perform correlation analysis
- Save cleaned datasets for future analysis

3.2 ASSIGNMENT OVERVIEW

The assignment focuses on understanding the cleaned Stack Overflow survey dataset through EDA techniques.

Main tasks include:

1. Understanding dataset structure
2. Visualizing variable distributions
3. Handling missing data
4. Identifying outliers
5. Computing statistics
6. Correlation analysis
7. Cross-tabulation analysis

8. Trend analysis for:
    - Job satisfaction
    - Remote work
    - Programming languages
    - Compensation

3.3 LABS

Lab 12: EXPLORATORY DATA ANALYSIS

Purpose of the Lab

This lab combines all major EDA tasks into one workflow.

The goal is to:

- Explore the dataset
- Understand variables
- Analyze trends
- Prepare data for visualization and modeling

Step 1: Load and Explore the Dataset

Common Functions Used

```
df.head()
df.info()
df.describe()
df.shape
df.columns
```

Purpose

These functions help:

- Understand dataset dimensions
- Identify column names
- Check data types

- View summary statistics
- Detect missing values

Step 2: Handling Missing Values

Missing values can distort analysis and visualizations.

Common Methods

Check Missing Values

```
df.isnull().sum()
```

Drop Missing Values

```
df.dropna()
```

Fill Missing Values

```
df.fillna()
```

Step 3: Analyze Key Variables

Key variables analyzed:

- Working hours
- Job satisfaction
- Remote work trends
- Programming language usage
- Education level
- Employment status

Step 4: Cross-Tabulation Analysis

Cross-tabulation helps compare categorical variables.

Example

```
pd.crosstab(df['Employment'], df['EdLevel'])
```

Purpose

Used to analyze relationships such as:

- Education vs employment type

- Remote work vs region
- Experience vs job satisfaction

Step 5: Save the Cleaned Dataset

Code

```
df.to_csv('cleaned_analyzed_dataset.csv', index=False)
```

Purpose

Exports cleaned dataset for:

- Future analysis
- Dashboard creation
- Sharing
- Reporting

Key Learnings from Lab 12

You learned how to:

- Explore dataset structure
- Handle missing data
- Analyze important variables
- Investigate language trends
- Study remote work patterns
- Perform cross-tabulation analysis

LAB 13: FINDING HOW THE DATA IS DISTRIBUTED

Purpose of Distribution Analysis

Distribution analysis helps understand:

- Data spread
- Central tendency
- Skewness

- Outliers
- Variable behavior

Common Distribution Visualizations

1. Histogram

Used to show frequency distribution.

Example

```
plt.hist(df['ConvertedCompYearly'])
```

Helps Identify

- Skewed data
- Peaks
- Spread
- Outliers

2. Box Plot

Used to visualize:

- Median
- Quartiles
- Outliers

Example

```
sns.boxplot(x=df['ConvertedCompYearly'])
```

Important Variables Examined

- Job satisfaction
- Compensation
- Programming languages
- Remote work
- Experience level

Comparing Languages

The lab compares:

- Languages respondents currently use
- Languages respondents want to learn

This helps identify:

- Emerging technologies
- Future trends
- Popular programming languages

Remote Work Analysis

Remote work preferences are analyzed by region.

Insights include:

- Remote work popularity
- Geographic work trends
- Workforce flexibility

Correlation Between Experience and Satisfaction

The relationship between:

- Years of professional coding experience
- Job satisfaction

is analyzed using statistical techniques and plots.

Exporting Cleaned Dataset

Code

```
df.to_csv('cleaned_survey_data.csv', index=False)
```

Key Learnings from Lab 13

You learned to:

- Understand variable distributions
- Detect patterns
- Visualize trends
- Analyze relationships

- Export cleaned data

LAB 14: FINDING OUTLIERS

What are Outliers?

Outliers are extreme values that differ significantly from the rest of the data.

They can:

- Distort analysis
- Affect averages
- Mislead models

Why Handle Outliers?

Removing outliers improves:

- Accuracy
- Reliability
- Statistical validity

Methods Used for Outlier Detection

1. Interquartile Range (IQR) Method

Formula

```
IQR = Q3 - Q1
```

Bounds

```
Lower Bound = Q1 - 1.5 × IQR
Upper Bound = Q3 + 1.5 × IQR
```

Values outside these bounds are considered outliers.

Example Process

Step 1: Calculate Quartiles

```
Q1 = df['ConvertedCompYearly'].quantile(0.25)
Q3 = df['ConvertedCompYearly'].quantile(0.75)
```

Step 2: Calculate IQR

```
IQR = Q3 - Q1
```

Step 3: Determine Bounds

```
lower = Q1 - 1.5 * IQR
upper = Q3 + 1.5 * IQR
```

Step 4: Remove Outliers

```
df_clean = df[(df['ConvertedCompYearly'] >= lower) &
              (df['ConvertedCompYearly'] <= upper)]
```

Converting Age to Numeric Values

Since age ranges are categorical, they are converted into numeric values.

Example Mapping

```
age_mapping = {
   "18-24 years old": 21,
   "25-34 years old": 29,
   "35-44 years old": 39
}
```

Purpose

Numeric values are required for:

- Correlation analysis
- Statistical calculations
- Heatmaps

Correlation Matrix

Code

```
correlation_matrix = numeric_df.corr()
```

Understanding Correlation Values

Correlation Value	Meaning
+1	Strong positive correlation
0	No correlation

Correlation Value	Meaning
-1	Strong negative correlation

Example Results

```
Age_numeric vs WorkExp = 0.852538
```

Meaning:

- Older respondents generally have more work experience.

Heatmap Visualization

Code

```
sns.heatmap(correlation_matrix, cmap='coolwarm')
```

Purpose

Visualizes relationships between variables.

Key Learnings from Lab 14

You learned to:

- Detect outliers
- Remove extreme values
- Compute correlations
- Create heatmaps
- Convert categorical age ranges into numeric format

LAB 15: FINDING CORRELATION

Purpose

This lab focuses specifically on understanding relationships between variables.

Correlation Analysis

Correlation helps answer questions such as:

- Does compensation increase with experience?
- Does job satisfaction affect salary?
- Are older developers paid more?

Scatter Plot Analysis

1. Compensation vs Work Experience

Code

```
sns.scatterplot(
    data=df_clean,
    x='WorkExp',
    y='ConvertedCompYearly'
)
```

Interpretation

This plot shows whether:

- Salary increases with experience
- Experience influences earnings

Observation

Generally:

- Higher experience leads to higher compensation.

2. Compensation vs Job Satisfaction

Code

```
sns.scatterplot(
    data=df_clean,
    x='JobSatPoints_1',
    y='ConvertedCompYearly'
)
```

Interpretation

Shows relationship between:

- Job satisfaction

- Annual compensation

Histograms and Boxplots

Used again to:

- Understand compensation distribution
- Detect skewness
- Identify remaining outliers

Heatmaps in Correlation Analysis

Heatmaps summarize all correlations visually.

Benefits

- Easy pattern recognition
- Faster interpretation
- Highlights strong relationships

Key Learnings from Lab 15

You learned how to:

- Perform correlation analysis
- Use scatter plots
- Analyze compensation trends
- Understand variable relationships
- Interpret correlation coefficients
- Use heatmaps effectively

IMPORTANT EDA TECHNIQUES LEARNED IN MODULE 3

Technique	Purpose
Histograms	Analyze distributions
Boxplots	Detect outliers

Technique	Purpose
Scatterplots	Study relationships
Heatmaps	Visualize correlations
Cross-tabulation	Compare categorical variables
Correlation analysis	Measure variable relationships
IQR Method	Remove outliers

OVERALL SUMMARY OF MODULE 3

In Module 3, you learned how to perform Exploratory Data Analysis (EDA) on the Stack Overflow Developer Survey dataset.

You explored:

- Dataset structure
- Missing values
- Data distributions
- Outlier detection
- Correlation analysis
- Trend analysis

You also created:

- Histograms
- Boxplots
- Scatterplots
- Heatmaps

These techniques help transform raw data into meaningful insights and prepare the dataset for dashboards, reporting, and predictive analysis

---

MODULE 4: DATA VISUALIZATION

4.1 MODULE OVERVIEW

Data Visualization is the process of representing data graphically to identify trends, patterns, relationships, and insights. In this module, the Stack Overflow Developer Survey dataset is visualized using different chart types.

4.1.1 Learning Objectives

By the end of this module, you should be able to:

- Create histograms to analyze data distributions.
- Create box plots to identify spread and outliers.
- Create scatter plots and bubble plots to analyze relationships.
- Create pie charts to visualize composition.
- Create stacked charts to compare multiple categories.
- Create line charts to identify trends over time or experience levels.
- Create bar charts to compare categorical values.
- Select the most appropriate visualization for a given analytical problem.

4.2 Labs

Lab 16: Data Visualization

Purpose

This lab serves as the foundation for all visualization activities in Module 4.

The dataset is stored in a relational database (SQLite), and SQL queries are used to extract data before visualization.

Example: Distribution of MainBranch

Count Categories

```
mainbranch_counts = (
    df['MainBranch']
    .value_counts()
)
```

Output

MainBranch	Count
Developer by profession	50,207
Not primarily developer but codes	6,511
Learning to code	3,875
Hobby coder	3,334
Former developer	1,510

Horizontal Bar Chart

```
sns.barplot(
    x=mainbranch_counts.values,
    y=mainbranch_counts.index
)
```

Interpretation

The majority of survey respondents are professional developers.

This visualization clearly compares respondent categories.

Visualization Techniques Covered

Distribution Analysis

- Histograms
- Box Plots

Relationship Analysis

- Scatter Plots
- Bubble Plots

Composition Analysis

- Pie Charts
- Stacked Charts

Comparison Analysis

- Line Charts
- Bar Charts

4.2.1 Visualizing Distribution of Data

Lab 17: Histograms

What is a Histogram?

A histogram displays the frequency distribution of numerical data.

Used For

- Understanding data spread
- Detecting skewness
- Identifying peaks
- Spotting unusual values

Example: Experience Groups

Create Experience Categories

```
bins = [0, 5, 10, 20, 50]
labels = [
    '0-5',
    '5-10',
    '10-20',
    '20+'
]
jobsat_df['ExperienceGroup'] = pd.cut(
```

```
    jobsat_df['YearsCodePro'],
    bins=bins,
    labels=labels
)
```

Histogram of Job Satisfaction

```
sns.histplot(
    data=jobsat_df,
    x='JobSat',
    hue='ExperienceGroup',
    bins=10,
    multiple='stack'
)
```

Interpretation

The histogram compares:

- Job Satisfaction
- Professional Coding Experience

Possible findings:

- Experienced developers may report higher satisfaction.
- Certain experience groups may dominate the survey.

Histogram Applications

Compensation Distribution

```
plt.hist(df['ConvertedCompYearly'])
```

Used to:

- Understand salary ranges
- Detect salary concentration

Coding Experience Distribution

```
plt.hist(df['YearsCodePro'])
```

Used to:

- Identify experience levels of respondents

Working Hours Distribution

```
plt.hist(df['WorkExp'])
```

Used to:

- Analyze workload patterns

Key Takeaways

Histograms reveal:

- Frequency
- Shape of distribution
- Skewness
- Central tendency

Lab 18: Box Plots

What is a Box Plot?

A box plot summarizes data using:

- Minimum
- Q1 (25th percentile)
- Median
- Q3 (75th percentile)
- Maximum
- Outliers

Example Preparation

Convert Job Satisfaction values:

```
years_jobsat_df['JobSatPoints_6'] = (
    years_jobsat_df['JobSatPoints_6']
    .astype(int)
)
```

Box Plot

```
sns.boxplot(
    data=years_jobsat_df,
    x='JobSatPoints_6',
    y='YearsCodePro'
)
```

Interpretation

Shows:

- Distribution of experience levels
- Median experience
- Experience variability
- Outliers

for each job satisfaction level.

Box Plot Applications

Compensation Distribution

```
sns.boxplot(y=df['ConvertedCompYearly'])
```

Used for:

- Salary spread
- Outlier detection

Age Distribution

```
sns.boxplot(y=df['Age_numeric'])
```

Used for:

- Understanding age demographics

Compensation by Employment Type

```
sns.boxplot(
    x='Employment',
```

```
    y='ConvertedCompYearly'
)
```

Used for:

- Comparing salaries across employment categories

Advantages

Box plots quickly show:

- Median
- Spread
- Skewness
- Outliers

4.2.2 Visualizing Relationships

Lab 19: Scatter Plots

What is a Scatter Plot?

A scatter plot visualizes the relationship between two numerical variables.

Each point represents one observation.

Data Preparation

Keep Top 5 Countries

```
top_countries = (
    country_df['Country']
    .value_counts()
    .head(5)
    .index
)
```

Scatter Plot

```
sns.scatterplot(
    data=country_df,
    x='Age_numeric',
    y='YearsCodePro',
```

```
    hue='Country',
    style='Age',
    alpha=0.6
)
```

Variables

Variable	Meaning
Age_numeric	Respondent age
YearsCodePro	Professional coding experience
Country	Country grouping

Interpretation

Can reveal:

- Positive correlation between age and experience.
- Country-wise differences.
- Clusters of respondents.

Common Uses

Compensation vs Experience

```
sns.scatterplot(
    x='WorkExp',
    y='ConvertedCompYearly'
)
```

Compensation vs Satisfaction

```
sns.scatterplot(
    x='JobSat',
    y='ConvertedCompYearly'
)
```

Insights

Scatter plots help identify:

- Correlations
- Clusters

- Trends
- Outliers

Lab 20: Bubble Plots

What is a Bubble Plot?

A bubble plot extends a scatter plot by introducing bubble size as a third variable.

Visual Dimensions

Element	Represents
X-axis	Variable 1
Y-axis	Variable 2
Bubble Size	Variable 3

Example

Count Admired Technologies

```
admired_counts = (
    filtered_df
    .groupby(
        ['Country',
         'LanguageAdmired']
    )
    .size()
    .reset_index(name='Frequency')
)
```

Bubble Plot

```
plt.scatter(
    admired_counts['Country'],
    admired_counts['LanguageAdmired'],
    s=admired_counts['Frequency']/5
)
```

Interpretation

Large bubbles indicate:

- Highly admired technologies

Small bubbles indicate:

- Less admired technologies

Example Findings

- JavaScript may dominate several countries.
- Python may have larger admiration bubbles.
- Certain technologies may be region-specific.

Advantages

Bubble plots display:

- Relationship
- Magnitude
- Frequency

simultaneously.

4.2.3 Visualizing Composition of Data

Lab 21: Pie Charts

What is a Pie Chart?

A pie chart displays parts of a whole.

Each slice represents a percentage.

Example

Top Desired Embedded Technologies

```
top_embedded = (
    embedded_exploded[
        'EmbeddedWantToWorkWith'
    ]
    .value_counts()
    .head(5)
)
```

Results

Technology	Count
Raspberry Pi	9,792
Arduino	6,482
GNU GCC	5,870
CMake	4,693
Cargo	4,567

Pie Chart

```
plt.pie(
    top_embedded.values,
    labels=top_embedded.index,
    autopct='%1.1f%%'
)
```

Interpretation

Shows percentage share of:

- Raspberry Pi
- Arduino
- GNU GCC
- CMake
- Cargo

among respondents.

Other Pie Chart Applications

Language Preferences

```
LanguageWantToWorkWith
```

AI Tool Usage

```
AISelect
```

Cloud Platforms

```
PlatformAdmired
```

Advantages

Pie charts are ideal when:

- Categories are few.
- Goal is composition analysis.

Lab 22: Stacked Charts

What is a Stacked Chart?

A stacked chart compares totals while showing internal composition.

Create Pivot Table

```
platform_employment_counts = (
    filtered_df
    .groupby(
        ['Employment',
         'PlatformAdmired']
    )
    .size()
    .unstack(fill_value=0)
)
```

Stacked Bar Chart

```
platform_employment_counts.plot(
    kind='bar',
    stacked=True
)
```

Interpretation

Shows:

Employment Type

- Full-time
- Freelancer
- Student
- Job seeker

versus

Preferred Platforms

- AWS
- Azure
- Google Cloud
- Cloudflare
- Firebase

Insights

Can identify:

- Most popular platform per employment category.
- Platform preference differences.

Advantages

Stacked charts display:

- Total counts
- Category composition
- Cross-category comparisons

simultaneously.

Visualizing Comparison of Data

Lab 23: Line Charts

What is a Line Chart?

A line chart shows trends across an ordered variable.

Example

Median Job Satisfaction by Experience

```
jobsat_trend = (
    jobsat_df
    .groupby('WorkExp')
    ['JobSatPoints_6']
    .median()
)
```

Line Plot

```
plt.plot(
    jobsat_trend['WorkExp'],
    jobsat_trend['JobSatPoints_6'],
    marker='o'
)
```

Interpretation

Shows how job satisfaction changes with:

- Work experience
- Career progression

Sample Trend

Experience	**Satisfaction**
0 Years	5
1 Year	10
2 Years	15
3 Years	20
4 Years	20

Other Uses

Compensation Trend

Age vs Compensation

Experience vs Salary

WorkExp vs Compensation

Advantages

Line charts reveal:

- Growth
- Decline
- Patterns
- Turning points

Lab 24: Bar Charts

What is a Bar Chart?

Bar charts compare values across categories.

Vertical Bar Chart

```
plt.bar(
    country_counts.index,
    country_counts.values
)
```

Horizontal Bar Chart

```
country_counts.sort_values().plot(
    kind='barh'
)
```

Example

Top 10 Countries by Respondent Count

Shows:

- Most represented countries
- Survey participation distribution

Additional Uses

Professional Roles

```
MainBranch
```

Desired Languages

```
LanguageWantToWorkWith
```

Popular Databases

```
DatabaseHaveWorkedWith
```

Compensation by Age Group

```
groupby('Age').median()
```

Advantages

Bar charts are best for:

- Category comparison
- Rankings
- Frequency analysis

Module 4 Important Visualization Selection Guide

Goal	Best Visualization
Distribution of one numeric variable	Histogram
Detect outliers	Box Plot
Relationship between 2 variables	Scatter Plot
Relationship between 3 variables	Bubble Plot
Percentage composition	Pie Chart
Composition + Comparison	Stacked Chart
Trend over time/experience	Line Chart
Compare categories	Bar Chart

MODULE 5: BUILDING A DASHBOARD

5.1 MODULE OVERVIEW

This module focuses on creating interactive dashboards using either:

- IBM Cognos Analytics
- Google Looker Studio

The goal is to transform the Stack Overflow Developer Survey data into meaningful visual dashboards that help stakeholders understand:

- Current technology usage
- Future technology trends
- Developer demographics

5.1.1 Learning Objectives

By the end of this module, you should be able to:

- Import datasets into IBM Cognos Analytics
- Create dashboards with multiple tabs
- Select appropriate chart types for different business questions
- Configure chart properties and labels
- Create interactive visualizations
- Export dashboards as PDF reports
- Present technology trends effectively

LAB 25: BUILDING A DASHBOARD WITH IBM COGNOS ANALYTICS

Purpose

Create a professional dashboard consisting of three tabs:

Dashboard Tabs

Tab	Purpose
Current Technology Usage	Technologies currently used by developers
Future Technology Trend	Technologies developers want to learn
Demographics	Developer population characteristics

Dataset Used

File

```
survey_data_updated.csv
```

This dataset contains cleaned Stack Overflow Developer Survey responses.

Dashboard Structure

Create a dashboard with:

Template

2 × 2 Rectangle Areas Tabbed Template

Each dashboard tab contains:

Panel 1
Panel 2
Panel 3
Panel 4

Total:

3 Tabs × 4 Visualizations = 12 Visualizations

Tab 1: Current Technology Usage

Objective

Understand technologies developers currently use.

Panel 1

Top 10 Programming Languages Used

Dataset Column

LanguageHaveWorkedWith

Chart Type

Bar Chart

Fields

Cognos Field	Value
Bars	Language
Length	Count
Color	Language

Features

- ✅ Show Value Labels
- ✅ Proper Title

Example Title

Top 10 Programming Languages Used

Why Bar Chart?

Bar charts are ideal for comparing frequencies across categories.

Panel 2

Top 10 Databases Used

Dataset Column

DatabaseHaveWorkedWith

Chart Type

Column Chart

Fields

Field	Value
Bars	Database
Length	Count
Color	Database

Features

- ✅ Show Value Labels
- ✅ Proper Title

Example Title

Top 10 Databases Used

Why Column Chart?

Makes it easy to compare database popularity.

Panel 3

Top 10 Platforms Used

Dataset Column

PlatformHaveWorkedWith

Chart Type

Word Cloud

Fields

Field	Value
Words	Platform
Size	Frequency
Color	Frequency

Features

✅ Proper Title

Example Title

Top Platforms Used by Developers

Why Word Cloud?

Frequently used platforms appear larger.

Quickly highlights dominant platforms.

Panel 4

Top 10 Web Frameworks Used

Dataset Column

```
WebframeHaveWorkedWith
```

Chart Type

Hierarchy Bubble Chart

Fields

Field	Value
Bubbles	Framework
Size	Frequency
Color	Frequency

Features

✅ Proper Title

Example Title

Top Web Frameworks Used

Why Hierarchy Bubble?

Displays popularity through bubble size.

Large bubble = widely used framework.

Tab 2: Future Technology Trend

Objective

Understand technologies developers want to learn or use in the future.

Panel 1

Top 10 Desired Languages

Dataset Column

```
LanguageWantToWorkWith
```

Chart Type

Bar Chart

Features

- Show Value Labels
- Proper Title

Example Title

Top 10 Languages Desired Next Year

Business Insight

Shows future language demand.

Useful for:

- Training programs
- Hiring strategies

- Skill development

Panel 2

Top 10 Desired Databases

Dataset Column

DatabaseWantToWorkWith

Chart Type

Column Chart

Features

- Show Value Labels
- Proper Title

Example Title

Top 10 Databases Desired Next Year

Business Insight

Helps predict future database trends.

Panel 3

Desired Platforms

Dataset Column

PlatformWantToWorkWith

Chart Type

Tree Map

Fields

Field	Value
Area Hierarchy	Platform
Size	Frequency

Field	**Value**
Heat	Frequency

Features

✅ Contrast Label Color

Example Title

Desired Platforms for Next Year

Why Tree Map?

Shows:

- Relative popularity
- Category proportions

using area size.

Panel 4

Desired Web Frameworks

Dataset Column

```
WebframeWantToWorkWith
```

Chart Type

Hierarchy Bubble Chart

Fields

Field	**Value**
Bubbles	Framework
Size	Frequency
Color	Frequency

Example Title

Top Web Frameworks Desired Next Year

Business Insight

Identifies emerging framework trends.

Tab 3: Demographics

Objective

Understand the characteristics of survey respondents.

Panel 1

Age Distribution

Dataset Column

Age

Chart Type

Pie Chart

Fields

Field	Value
Segments	Age Group
Size	Count

Features

✅ Display %

Example Title

Respondent Distribution by Age

Why Pie Chart?

Shows percentage share of each age group.

Panel 2

Respondent Count by Country

Dataset Column

Country

Chart Type

Map Chart

Fields

Field	Value
Region Location	Country
Region Color	Count

Example Title

Respondent Distribution by Country

Why Map Chart?

Provides geographic visualization of respondents.

Panel 3

Education Level Distribution

Dataset Column

EdLevel

Chart Type

Line Chart

Fields

Field	Value
X-axis	Education Level
Y-axis	Count

Features

- [x] Show Value Labels
- [x] Show Markers

Example Title

Respondent Distribution by Education Level

Why Line Chart?

Shows trends across education categories.

Panel 4

Age vs Education Level

Dataset Columns

```
Age
EdLevel
```

Chart Type

Stacked Bar Chart

Fields

Field	Value
Bars	Age
Length	Count
Color	Education Level

Features

✅ Show Value Labels

Example Title

Respondent Count by Age and Education Level

Why Stacked Bar Chart?

Displays:

- Age distribution
- Educational composition

simultaneously.

DASHBOARD CREATION WORKFLOW

Step 1

Upload:

```
survey_data_updated.csv
```

into Cognos Analytics.

Step 2

Create Dashboard

Choose:

2×2 Rectangle Template

Step 3

Create Tab 1

Rename:

Current Technology Usage

Step 4

Create Tab 2

Rename:

Future Technology Trend

Step 5

Create Tab 3

Rename:

Demographics

Step 6

Add Required Visualizations

Ensure:

- Correct chart type
- Correct fields
- Proper titles
- Value labels enabled where required

Step 7

Export Dashboard

Click:

Share → Export

Choose:

Landscape Orientation

Then:

Save as PDF

Step 8

Take Screenshots

Required for:

- Final presentation
- Peer review grading
- Portfolio documentation

KEY DASHBOARD CONCEPTS

Current Technology Usage

Measures technologies currently used by developers.

Columns:

- LanguageHaveWorkedWith
- DatabaseHaveWorkedWith
- PlatformHaveWorkedWith
- WebframeHaveWorkedWith

Future Technology Trend

Measures technologies developers want to learn.

Columns:

- LanguageWantToWorkWith
- DatabaseWantToWorkWith
- PlatformWantToWorkWith
- WebframeWantToWorkWith

Demographics

Measures respondent characteristics.

Columns:

- Age
- Country
- EdLevel

QUICK REVISION TABLE

Dashboard Tab	Visualization Types
Current Technology Usage	Bar, Column, Word Cloud, Hierarchy Bubble
Future Technology Trend	Bar, Column, Tree Map, Hierarchy Bubble
Demographics	Pie, Map, Line, Stacked Bar

MODULE 6: FINAL ASSIGNMENT – PRESENT YOUR FINDINGS

6.1 IMPORTANCE OF PRESENTING FINDINGS

Data analysis does not end after collecting, cleaning, and analyzing data. The final and most important step is communicating insights effectively to stakeholders.

A well-presented report helps:

- Transform data into actionable business decisions
- Communicate findings clearly
- Highlight key trends and recommendations
- Support strategic planning
- Demonstrate analytical skills professionally

In this Capstone Project, the final deliverable is a professional PowerPoint presentation summarizing all analyses, dashboards, and findings derived from the Stack Overflow Developer Survey dataset.

6.2WHAT IS A FINDINGS REPORT?

A Findings Report is a structured document that presents:

- Results of analysis
- Key insights
- Methodology used
- Conclusions
- Recommendations

Its purpose is to answer business questions using evidence gathered from data.

Example

Business Question:

Which programming languages are most in demand?

Finding:

Python, JavaScript, and SQL are currently the most widely used technologies among developers.

6.3 WHY FINDINGS REPORTS MATTER

Without proper reporting:

- Valuable insights may be overlooked

- Decision-makers may not understand findings
- Analysis loses business value

Findings report bridges the gap between:

Data Analysts

Who understands the data

Stakeholders

Who need actionable information

6.4 KEY ELEMENTS OF A SUCCESSFUL FINDINGS REPORT

The course highlights the following essential sections:

A. Cover Page

The first page of the report.

Include:

- Report title
- Your name
- Organization/Institution
- Date
- Contact information (optional)

Example

Technology Trends Analysis Using Stack Overflow Developer Survey 2025

Prepared by: Chavi Jain

IBM Data Analyst Capstone Project

June 2026

B. Executive Summary

A concise overview of:

- Purpose
- Key findings

- Recommendations

Usually:

- 1 page
- Written after completing the report

Example

This study analyzed developer technology preferences using the Stack Overflow Survey dataset. Python, JavaScript, and SQL emerged as the most widely used technologies. PostgreSQL showed the strongest future growth trend. Cloud platforms such as AWS and Azure remain dominant among professional developers.

C. Table of Contents

Helps readers navigate the report.

Example:

1. Introduction
2. Methodology
3. Results
4. Discussion
5. Conclusion
6. References
7. Appendix

D. Introduction

Explains:

- Business problem
- Objectives
- Research questions

Example

The purpose of this analysis is to identify current and future technology trends using the Stack Overflow Developer Survey dataset.

Research Questions:

- Which programming languages are most used?
- Which databases are most desired?
- Which platforms are growing?
- What skills should organizations prioritize?

E. Methodology

Describes:

Data Sources

- Stack Overflow Survey
- APIs
- Web Scraping
- Job posting datasets

Methods Used

- Data Collection
- Data Wrangling
- Exploratory Data Analysis
- Visualization
- Dashboard Creation

Tools Used

- Python
- Pandas
- NumPy
- Matplotlib

- Seaborn
- SQL
- SQLite
- IBM Cognos Analytics
- PowerPoint

F. Results

The most important section.

Shows:

- Findings
- Charts
- Graphs
- Dashboard screenshots

Examples:

Current Technology Usage

Top Languages:

- Python
- JavaScript
- SQL
- Java
-

Top Databases

- PostgreSQL
- MySQL
- MongoDB

Top Platforms

- AWS
- Azure

- Google Cloud

G. Discussion

Explains:

What do the results mean?

Example

Python remains dominant because of its applications in:

- Data Science
- Machine Learning
- AI
- Automation

Organizations investing in Python training are likely to benefit from increasing industry demand.

H. Conclusion

Summarizes:

- Major findings
- Recommendations
- Future trends

Example

The analysis indicates growing adoption of cloud technologies, AI-related tools, and modern databases. Python, PostgreSQL, and AWS are expected to remain highly valuable skills in the coming years.

I. Appendix

Contains supporting material such as:

- Additional charts
- SQL queries
- Code snippets

- Data dictionaries
- Dashboard screenshots

6.5 REPORT STRUCTURE FROM IBM DATA SCIENCE TEXTBOOK

Recommended structure:

1. Cover Page
2. Table of Contents
3. Executive Summary
4. Introduction
5. Literature Review (optional)
6. Methodology
7. Results
8. Discussion
9. Conclusion
10. References
11. Acknowledgements
12. Appendix

6.6 BEST PRACTICES FOR PRESENTING FINDINGS

Practice 1: Make Visuals Readable

Ensure:

- Large font sizes
- Clear labels
- Visible legends
- Proper spacing

Avoid

❌ Tiny charts

❌ Small labels

❌ Crowded slides

Practice 2: Use Data as Supporting Evidence

Do not overwhelm audiences with numbers.

Instead:

1. Present the insight
2. Use data to support it

Example:

Insight:

Python is the fastest-growing language.

Supporting Data:

42% of respondents want to learn Python next year.

Practice 3: Tell a Story

Good presentations follow a logical narrative:

Problem → Analysis → Findings → Recommendations

Example:

Question: Which technologies should companies invest in? ↓ Analysis: Survey of 60,000+ developers ↓ Finding: Python and Cloud Platforms dominate ↓

Recommendation:
Focus on Python, AWS, and PostgreSQL training

Practice 4: One Message per Visualization

Each chart should communicate a single idea.

Bad Example

A chart showing:

- Languages
- Databases
- Salaries
- Countries

all together

Good Example

One chart:

Top 10 Programming Languages

Practice 5: Remove Unnecessary Data

Only include information that supports the story

Avoid:

- Extra metrics
- Irrelevant charts
- Excessive tables

Practice 6: Highlight Key Insights

Use:

- Titles
- Annotations

- Callouts

Example:

"PostgreSQL shows the highest future adoption among databases."

6.7 POWERPOINT FOR THE WEB

Microsoft provides a free online version of PowerPoint.

Getting Started with PowerPoint for Web

Step 1

Sign in to Microsoft account

Step 2

Open PowerPoint

Step 3

Create:

- New Blank Presentation

or

- Use Template

Basic PowerPoint Skills

Students learned how to:

Create Presentations

- Blank Presentation
- Template-Based Presentation

Manage Slides

- Add Slide
- Delete Slide
- Reorder Slides

Change Layouts

Examples:

- Title Slide
- Title and Content
- Picture with Caption

Design Slides

Using:

- Designer Pane
- Themes
- Templates

Adding Content to Slides

Students practiced adding:

Text

- Titles
- Bullet Points
- Summaries

Images

- Search Online
- Insert Pictures

Videos

- Embed videos when necessary

Icons and Art

- Improve visual appeal
